UNIT 7 · STANDARD 6.EE.5

Solve and Graph Inequalities

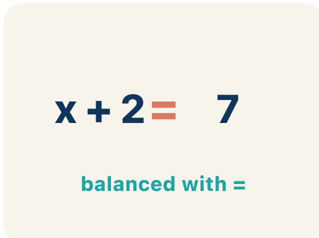
Lesson 7-6

Name: ____________________ Date: ____________________ Class: ____________________

Key Vocabulary

Level 1 support

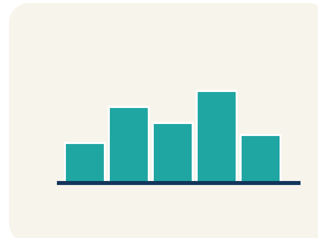
Picture first, then the word, then a plain-language meaning. Say each word out loud.



$x + 3 > 10$ → subtract 3 → $x > 7$ — any number greater than 7 works

Solve

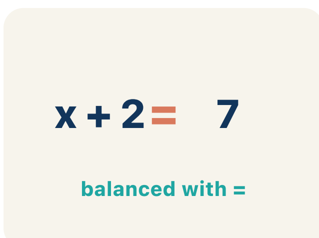
To find the number that makes it true.



$x > 7$: open circle at 7, shade right — shows all solutions at a glance

Graph

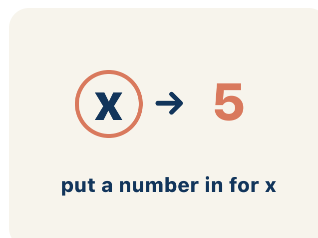
To show answers on a number line with circles and shading.



$x = 8$ is a solution to $x > 7$ because $8 > 7$ is true; $x = 5$ is not because $5 > 7$ is false

Solution

A number that makes the equation or inequality true.



Substitute $x = 5$ into $x + 3 > 10$: $5 + 3 = 8$, and $8 > 10$ is false, so $x = 5$ is not a solution

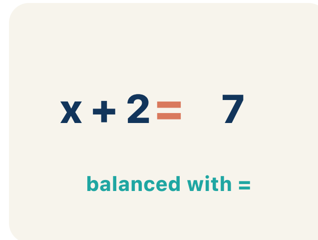
Substitute

To put a number in for the letter to check if it works.

For $x > 4$, the solution set is every number greater than 4, such as 5, 6, 7, ...

Solution set

All of the values that make an inequality true.



To solve $x + 3 > 7$, use the inverse of adding 3: subtract 3 to get $x > 4$.

Inverse operation

An operation that undoes another, used to isolate the variable.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can solve an inequality and graph its solution set on a number line.

1 Notice

Detective Kim knows that the suspect withdrew more than $50 from an ATM.

2 Key idea

I can solve an inequality and graph its solution set on a number line.

3 Words I'll use

Solve **Graph** **Solution** **Substitute** **Solution set** **Inverse operation**

🤔 Think About It

- What is the unknown in this situation?
- What operation do you need to undo?
- Will the answer be one number or many numbers?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: Solve: $x + 6 > 14$

A. $x > 8$

B. $x > 20$

C. $x < 8$

D. $x = 8$

Step 1 Subtract 6 from both sides: $x > 14 - 6 = 8$.

Step 2 Graph: open circle at 8, shade right.

✅ **Answer:** A. $x > 8$

🤝 We do — together

Solve this one with your class using the same steps.

Problem: Solve: $x - 9 \le 3$

A. $x \le 12$

B. $x \le 6$

C. $x \ge 12$

D. $x = 12$

Step 1 ______________________________

Step 2 ______________________________

Answer: ______________________________

✏️ You do — your turn

Now try one on your own. Show every step.

Problem: Solve: $x + 10 \ge 25$

A. $x \ge 15$

B. $x \ge 35$

C. $x \le 15$

D. $x = 15$

Show your work:

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Solving an inequality and solving an equation both isolate the variable. What is the same about your steps, and what is different about the answer you get?

Level 1 support

Start here: Solving an inequality gives a range of solutions, not one number.

💡 **Need a hint?**

Sentence starters (optional)

The steps are the same because I ____ to ____ the variable.

Los pasos son iguales porque ____ para ____ la variable.

The answer is different because an inequality has ____.

La respuesta es diferente porque una desigualdad tiene ____.

Word bank: solve | graph | solution | substitute

Listen for: *Students note both use inverse operations to isolate the variable, but an inequality's solution is many values (a range) while an equation's is a single value.*

Level 2

Predict: after you solve an inequality, how will the graph look different from graphing the solution to an equation? Justify.

- The inequality graph shows ____ while an equation shows ____.
- I would shade because ____.

EXPLORE

What steps did you follow to solve and then graph the inequality, and how did you check a value to make sure your solution set is right?

Level 1 support

Start here: I solve to isolate the variable, then substitute a value to test the solution set.

💡 **Need a hint?**

Sentence starters (optional)

First I solved by ____, then I graphed ____.

Primero resolví al ____, luego grafiqué ____.

I substituted ____ to check that it ____ the inequality.

Sustituí ____ para verificar que ____ la desigualdad.

Word bank: **solve** **graph** **solution** **substitute**

Listen for: *Students isolate the variable, graph the solution set with the correct circle and shading, and substitute a test value to confirm it makes the inequality true.*

Level 2

If you substitute a value from the shaded region and it makes the inequality false, what does that tell you about your solution? Explain.

- If a shaded value makes it false, then I ____.
- A correct solution set means every shaded value ____.

CONNECT

When would solving an inequality help you stay within a real-life limit?

Level 1 support

Start here: Inequalities describe rules where many amounts are allowed up to a limit.

💡 **Need a hint?**

Sentence starters (optional)

Solving an inequality would help me stay within ____ when ____.

Resolver una desigualdad me ayudaría a quedarme dentro de ____ cuando ____.

I would substitute ____ to check I am within the limit.

Sustituiría ____ para verificar que estoy dentro del límite.

Word bank: solve | graph | solution | substitute

Listen for: *Students give a real budget/time/capacity limit and explain that the inequality's solution set shows all amounts that keep them within the limit.*

Level 2

Critique: 'The solution to an inequality is just like the solution to an equation — one number.' Use a solved example to explain why this is wrong.

- This is wrong because an inequality solution is ____.
- For example, solving ____ gives all values that ____.

Write About the Math
The Writing Revolution

I can explain my work using the words solve, graph, solution, and substitute.

1. Kernel Sentence
subject + verb

Model: Solve is to find the number that makes it true.
Resolver es encontrar el número que la hace verdadera.

Write a kernel sentence about solve. Use a subject and a verb.
Escribe una oración base sobre resolver. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion
because · but · so

Kernel: Solve matters in math
Resolver importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Solve matters in math because ____.** *Resolver importa en matemáticas porque ____.* ______________________________
but *pero*	**Solve matters in math, but ____.** *Resolver importa en matemáticas, pero ____.* ______________________________
so *entonces*	**Solve matters in math, so ____.** *Resolver importa en matemáticas, entonces ____.* ______________________________

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about solve.
Di un hecho verdadero sobre solve.

Solve ____.

Question
Pregunta

Ask a question about solve.
Haz una pregunta sobre solve.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about solve.
Muestra entusiasmo sobre solve.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with solve.
Dile a un compañero qué hacer con solve.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

The steps are the same because I ____ to ____ the variable.
Los pasos son iguales porque ____ para ____ la variable.

The answer is different because an inequality has ____.
La respuesta es diferente porque una desigualdad tiene ____.

First I solved by ____, then I graphed ____.
Primero resolví al ____, luego grafiqué ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A rider must be at least 48 inches tall. Which inequality and a valid height fit?

A. $h \geq 48$; 50 inches works

B. $h > 48$; 48 works

C. $h \leq 48$; 50 works

D. $h < 48$; 48 works

Show your work:

2. Which value is NOT a solution of $x \leq 7$?

A. 8

B. 7

C. 6

D. 0

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A detective has a budget of at most $100 for supplies. She already spent $37 on gloves. Write an inequality for the remaining amount r she can spend, solve it, graph it, and name three possible values for r.

Sentence starter: Inequality: 37 + r ≤ ____. Solving: r ≤ ____ − 37 = ____. Graph: ____ circle at ____, shade ____. Three possible values: ____.

Show your work:

____

____

____

____

Reflect — Exit Ticket

Solve and describe the graph: $x + 5 < 11$

A. $x < 6$; open circle at 6, shade left

B. $x < 16$; open circle at 16, shade left

C. $x \le 6$; closed circle at 6, shade left

D. $x > 6$; open circle at 6, shade right

Your answer:

____

____

____

Answer Key & Teacher Guide

1. **We Try (notes):** A. $x \le 12$ *— Add 9 to both sides: $x \le 3 + 9 = 12$. Graph: closed circle at 12, shade left.*
2. **You Try (notes):** A. $x \ge 15$ *— Subtract 10 from both sides: $x \ge 25 - 10 = 15$. Graph: closed circle at 15, shade right.*
3. **Try It 1:** A. $h \ge 48$; 50 inches works *— 'At least 48' is $h \ge 48$, and 50 satisfies it.*
4. **Try It 2:** A. 8 *— 8 is greater than 7, so it is not a solution.*
5. **Exit Ticket:** A. $x < 6$; open circle at 6, shade left *— Subtract 5 from both sides: $x < 11 - 5 = 6$. Open circle at 6 (not included), shade left. ✓*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Solve is to find the number that makes it true.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).